

MULTAN INTL, Pakistan

WMO: 416750

Lat: 30.203N

Lon: 71.419E

Elev: 400

StdP: 14.48

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.1	42.7	29.8	24.2	58.5	33.6	28.6	58.1	11.8	60.7	10.1	61.0	0.2	0	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	19.7	111.3	74.4	108.6	75.5	105.9	76.4	84.4	97.8	83.7	97.2	83.1	96.6	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.5	166.0	91.3	80.7	161.6	91.3	79.8	156.9	91.7	49.2	98.0	48.4	97.2	47.8	97.5	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.5	15.1	12.1	DB	36.8	116.0	2.6	2.0	34.9	117.5	33.4	118.6	31.9	119.7	30.0	121.1
			WB	34.6	86.8	3.0	1.2	32.5	87.6	30.7	88.3	29.1	89.0	26.9	89.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.7	55.6	61.9	72.3	83.8	93.1	95.5	93.7	91.1	88.4	81.0	68.8	58.6	(6)
	DBStd	14.76	4.17	4.90	6.31	6.30	4.98	4.44	3.59	2.97	3.41	4.79	4.13	4.49	(7)
	HDD50	11	8	0	0	0	0	0	0	0	0	0	0	3	(8)
	HDD65	628	292	110	12	0	0	0	0	0	0	0	11	203	(9)
	CDD50	10491	181	332	692	1013	1337	1364	1354	1273	1151	962	563	269	(10)
	CDD65	5632	0	22	239	563	872	914	889	808	701	497	123	4	(11)
	CDH74	83039	7	216	2394	7565	14066	15239	14462	12389	10063	5538	1029	71	(12)
	CDH80	53266	0	26	998	4552	9773	10954	10020	7950	5930	2820	241	2	(13)
(14) Wind	WSAvg	4.1	1.7	2.8	3.7	4.7	5.4	6.9	6.7	6.6	4.9	2.5	1.4	1.3	(14)
(15) Precipitation	PrecAvg	7.1	0.3	0.4	0.7	0.5	0.4	0.6	2.0	1.3	0.5	0.1	0.1	0.2	(15)
	PrecMax	14.2	1.7	3.7	2.4	2.6	2.7	3.2	9.1	6.8	6.2	0.9	1.1	4.3	(16)
	PrecMin	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	3.0	0.4	0.7	0.7	0.7	0.6	0.9	1.8	1.5	1.0	0.2	0.2	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	82.8	97.6	109.8	114.9	115.1	108.9	104.1	102.6	99.0	88.7	79.1	(19)
		MCWB	57.9	63.3	69.7	71.0	74.6	74.9	81.8	81.3	80.3	74.2	67.1	61.2	(20)
	2%	DB	71.9	79.4	93.1	105.9	111.4	111.7	105.8	101.8	100.4	96.9	85.6	76.1	(21)
		MCWB	56.5	62.0	69.4	70.3	73.1	76.1	82.0	81.6	79.0	74.7	66.5	59.9	(22)
	5%	DB	69.7	76.6	89.4	102.2	109.0	108.9	103.9	100.2	98.7	94.9	82.8	73.4	(23)
		MCWB	55.9	61.0	67.9	70.0	73.2	77.6	82.1	81.5	78.6	73.4	64.9	58.4	(24)
	10%	DB	67.1	73.4	85.8	98.2	105.9	105.9	101.9	98.5	96.8	92.4	80.3	70.4	(25)
		MCWB	54.5	59.2	66.9	69.4	73.1	78.6	82.0	81.6	78.8	72.5	64.2	57.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.4	66.9	73.7	74.8	81.0	84.2	85.7	85.3	84.2	79.2	70.9	64.0	(27)
		MCDB	70.2	77.0	91.5	100.4	100.9	101.4	99.6	95.8	96.2	92.7	82.6	74.7	(28)
	2%	WB	58.7	64.8	71.7	73.2	79.1	83.2	84.5	84.3	83.2	77.9	69.3	61.9	(29)
		MCDB	68.4	75.8	88.0	97.2	100.3	101.1	98.8	96.0	94.6	91.1	80.5	72.2	(30)
	5%	WB	57.4	63.0	69.9	72.0	77.5	82.3	83.8	83.5	82.1	76.6	67.6	60.1	(31)
		MCDB	66.4	73.7	84.6	94.5	99.6	100.9	98.1	95.2	93.3	89.7	78.1	70.0	(32)
	10%	WB	56.0	61.1	68.2	70.8	75.6	81.4	83.3	82.9	80.9	75.0	66.1	58.4	(33)
		MCDB	64.4	70.5	82.6	92.6	99.3	99.2	97.4	94.4	92.5	88.1	76.9	67.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.4	21.8	22.1	24.8	23.8	19.7	15.3	14.4	16.6	22.0	23.4	23.2	(35)
		MCDBR	25.8	25.5	25.8	28.6	27.3	24.2	17.0	16.0	18.3	22.5	25.7	26.8	(36)
	5% WB	MCWBR	12.6	10.8	8.8	7.8	7.6	7.0	5.1	5.1	5.7	6.9	9.8	12.9	(37)
		MCWBR	22.0	22.1	24.0	25.7	22.4	17.5	15.4	14.6	15.6	19.3	21.7	23.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.502	0.504	0.526	0.595	0.648	0.756	0.818	0.732	0.624	0.622	0.616	0.553	(40)
		taud	1.819	1.868	1.847	1.735	1.650	1.509	1.441	1.578	1.731	1.646	1.619	1.725	(41)
	Ebn at Noon		219	236	242	231	220	196	184	198	214	200	184	194	(42)
		Edh at Noon	56	58	64	74	81	93	99	85	71	72	69	59	(43)
(44) All-Sky Solar Radiation	RadAvg		1006	1324	1724	2060	2226	2135	1903	1846	1764	1517	1154	999	(44)
	RadStd		83	99	91	73	60	85	95	94	88	62	72	50	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page